# Paleomagnetism of miscellaneous Franklin and Mackenzie Diabases of the Canadian Shield, and their adjacent Country Rocks[1]

J. K. PARK

*Earth Physics Branch, Department of Energy, Mines, and Resources, Ottawa, Canada*

Received February 5, 1974

Revision accepted for publication April 18, 1974

This paper brings together the paleomagnetic results from miscellaneous unmetamorphosed diabases and their remagnetized adjacent country rocks collected from widely separated locations within the Canadian Shield. Most of these diabases are tentatively assigned to the Franklin and Mackenzie Igneous Episodes on the basis of their paleomagnetic poles and their geological relationships with previously assigned diabases. The tentative Mackenzie pole from the present work is 02.5°N, 169.8°W ($N = 12$, $K = 35$, $A_{95} = 7.4$).

Cet article réuni les résultats paléomagnétiques de divers diabases non-métamorphisés et de roches contiguës remagnétizés receuillis de lieux espacés dans le bouclier canadien. Le plupart de ces diabases sont attribués tentativement à base de leur pôles paléomagnétiques et leur relations géologiques avec des diabases déjà attribués aux épisodes ignés de Franklin et Mackenzie. Le pôle d'essai de Mackenzie du travail présent est de 02.5°N, 169.8°W ($N = 12$, $K = 35$, $A_{95} = 7.4$). [Traduit par le journal]

## Introduction

Extensive paleomagnetic surveys are now being undertaken in remote parts of Canada by several institutions. Small collections are often made in these areas from rock units that are not of immediate interest, but which are made because one can conveniently obtain samples. In this note, results from some of these miscellaneous collections are brought together and described. The samples are all from unmetamorphosed diabase dikes or sheets and from country rocks close to the contacts. Site localities are shown in Figs. 1 and 2 (also in Park 1973), and described along with other details in the Appendix. The emplacement of these bodies post-dates the regional metamorphism of the basement terrain. The purpose in sampling from the country rock was to determine whether the magnetization of the diabase was acquired at the time of cooling by using the classical contact test of Brunhes (1906).

## Methods

Three or more cores were drilled from each diabase body. A similar number were obtained from the adjacent country rocks (where these could be sampled). Two specimens were cut from each core. The sample (mean core) data serve as the basic units in the statistical analysis. Specimens were cleaned in alternating fields (a.f.) using the apparatus described by Roy *et al.* (1969, 1973) and measured with commercial spinner magnetometers.

Results from samples have been discarded if (1) the difference between the magnetic directions of two specimens from the same core exceeds 45°; (2) the coercivity (median destructive field of n.r.m.) is less than 50 oe; or (3) the intensities relative to those of other samples in the site are very high and the directions scattered—a characteristic of remagnetization produced by lightning (Cox 1961; Graham 1961). All acceptable samples show end-points, though some exhibit randomly-directed temporary components in the range of the cleaning field. Results from several complete sites are also considered unreliable. These

[1]Contributions from the Earth Physics Branch No. 485.

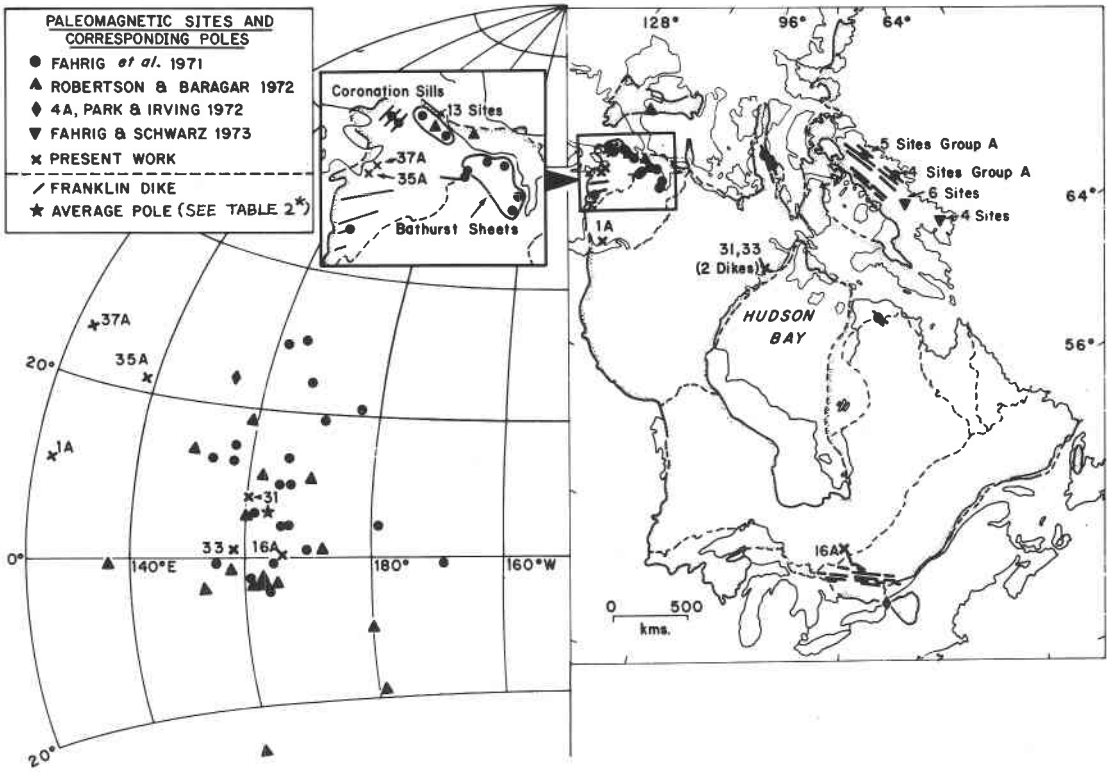



FIG. 1. Franklin paleomagnetic sites and their corresponding poles. Several unassigned diabase sites are also shown. The exposed land extent and probable water extent of the Canadian Shield are respectively shown by heavy solid lines and dashed lines—both rimmed with shading—and the structural provinces within the shield are indicated by lighter dashed lines.

are sites which exhibit low coercivities ($35C_1$, $35C_2$, 39C, 86D) and low intensities (1AC: $0.2 \times 10^{-6}$ emu/cc). These sites are listed, but are not used for calculating any averages (Table 1).

With the elimination of the unstable and unreliable samples and sites, a number of the remaining diabases are seen to have directions which noticeably differ from their adjacent country rocks (up to 30°, Table 1). This phenomenon has been found by other workers (*e.g.*, Irving *et al.* 1972*a*, *b*). In the present case, it is unlikely to be the result of residual components of the original remanence of the country rock; since most of the samples from each site have been collected from within one meter (and often at the margin) of diabases which range in thickness from 15 to 85 m, and which samples are considered to lie within the zone of complete remagnetization. Since the reason for these discrepancies is not known—there being no firm theory to predict the degree of magnetic interaction between the diabases and the remagnetized country rocks, or to indicate which, if either, gives the correct direction of the earth's field at the time of diabasic intrusion—the two have been simply meaned together using sample unit weight.

## Diabase Assignment

The diabases studied herein have poles which are generally similar to those of either the Franklin (Fahrig *et al.* 1971; Robertson and Baragar 1972; Fahrig and Schwarz 1973) or the Mackenzie diabases (Fahrig *et al.* 1965; Larochelle 1967, Fahrig and Jones 1969; Robertson 1969; Irving *et al.* 1972*b*; Tables 1 and 2). Since the mean pole positions of the

TABLE 1. Summary of directions from Franklin and Mackenzie sites

| Site | Lat(N°), Long(W°) | *N* | a.f. (oe) | *D*°, *I*° (*R*) | Pole |
|---|---|---|---|---|---|
| | Diabase and 'contacts' assigned to the Mackenzie igneous episode | | | | |
| 22D | 64.1, 082.4 | 5 | 250 | 269, +07 (4.95) | 03N, 170W |
| 22C | 64.1, 082.4 | 4(1) | 750 | 258, +03 (3.99) | 04S, 161W |
| **22 Mean** | — | 9 | — | 264, +05 (8.90) | **01S, 166W** |
| **24D** | 64.1, 082.4 | 5 | 300 | 261, +13 (4.94) | **02N, 162W** |
| **35D** | 64.4, 089.3 | 11 | 400 | 267, +18(10.86) | **07N, 172W** |
| 35C1 | 64.4, 089.3 | 5 | 250 | 264, +42 (4.70) | 19N, 163W |
| 35C2 | 64.4, 089.3 | 3(2) | 200 | 268, +41 (2.64) | 20N, 166W |
| 38D | 67.7, 091.1 | 10 | 250 | 267, +16 (9.95) | 06N, 175W |
| 38C | 67.7, 091.1 | 5 | 500 | 266, +26 (4.99) | 11N, 172W |
| **38 Mean** | — | 15 | — | 267, +19(14.88) | **08N, 174W** |
| **39D** | 66.4, 087.7 | 8 | 500 | 258, +21 (7.85) | **05N, 172W** |
| 39C | 66.4, 087.7 | 4(1) | 150 | 299, +74 (3.58) | 63N, 160W |
| 5AD | 62.2, 109.7 | 5 | 300 | 247, +27 (4.97) | 02N, 173W |
| 5AC | 62.2, 109.7 | 3 | 400 | 272, +46 (2.93) | 25N, 187W |
| **5A Mean** | — | 8 | — | 255, +34 (7.68) | **10N, 177W** |
| **10AD** | 62.9, 113.3 | 5 | 400 | 218, +35 (4.99) | **02S, 149W** |
| 13AS | 62.2, 111.4 | 5(2) | 400 | 236, +07 (4.96) | 12S, 169W |
| 13AC | 62.2, 111.4 | 5 | 500 | 248, +34 (4.82) | 07N, 173W |
| **13A Mean** | — | 10 | — | 241, +20 (9.46) | **03S, 171W** |
| 84D | 62.0, 109.6 | 6(1) | 250 | 272, +27 (5.90) | 14N, 195W |
| 84C | 62.0, 109.6 | 5 | 500 | 243, +20 (4.96) | 03S, 171W |
| **84 Mean** | — | 11 | — | 258, +25(10.56) | **06N, 183W** |
| 86D | 62.0, 109.6 | 5(3) | 300 | 292, +55 (4.89) | 41N, 199W |
| **86C** | 62.0, 109.6 | 2(1) | 500 | 248, +19 (1.96) | **01S, 176W** |
| **103S** | 62.7, 110.5 | 5 | 300 | 246, +31 (4.98) | **05N, 172W** |
| **104S** | 62.8, 110.4 | 5 | 400 | 242, +14 (4.99) | **06S, 173W** |
| | Diabase and 'contacts' assigned to the Franklin igneous episode | | | | |
| 31D | 64.1, 089.8 | 8(2) | 500 | 115, −03 (7.43) | 12N, 158E |
| 31C | 64.1, 089.8 | 5 | 200 | 107, +11 (4.90) | 02N, 162E |
| **31 Mean** | — | 13 | — | 111, +03(12.19) | **08N, 160E** |
| 33D | 64.1, 089.8 | 6(4) | 200 | 115, +08 (5.85) | 07N, 155E |
| 33C | 64.1, 089.8 | 5 | 400 | 104, +28 (4.75) | 08S, 161E |
| **33 Mean** | — | 11 | — | 111, +17(10.38) | **01N, 158E** |
| 16AD | 47.3, 079.8 | 5 | 400 | 285, −34 (4.96) | 04S, 167E |
| 16AC | 47.3, 079.8 | 3 | 250 | 295, −24 (2.99) | 07N, 163E |
| **16 Mean** | — | 8 | — | 289, −31 (7.90) | **000, 166E** |
| | Unassigned diabases | | | | |
| 1AD | 62.2, 112.3 | 4(1) | 500 | 306, −10 (3.90) | 11N, 123E |
| 1AC | 62.2, 112.3 | 1(1) | 100 | 274, −36 (—) | — |
| 35AS | 65.5, 117.3 | 6 | 200 | 292, +27 (5.86) | 22N, 138E |
| 37AS | 65.6, 116.9 | 4(1) | 200 | 307, +24 (3.54) | 26N, 123E |

NOTES: D, S, and C after site numbers indicate dike, sill, and adjacent country rock respectively. Sites used in the grand average are in boldface. *N* is the number of statistical units used in the analysis; samples (either cores or hand samples) being given unit weight within each site. Bracketed numbers after *N* denote the number of samples variously discarded. Column 4 indicates the cleaning field in oersteds. *D*°, *I*° are the declination and inclination (with respect to horizontal and true north) of the mean site directions after cleaning, and *R* (in brackets) is the resultant of *N* unit vectors.

Franklin and Mackenzie are roughly similar, though significantly different (Figs. 1 and 2), a problem arises in assigning the present diabases to one of these groups. One method is by obtaining accurate age determinations at all sites. This is obviously the best method in principle, but unfortunately results are available from only two sites and these are of a preliminary nature (see appendix). The method used here for distinguishing between the two diabase groups involved establishing geological relationships between present and earlier paleomagnetic sampling sites (see appendix for details).

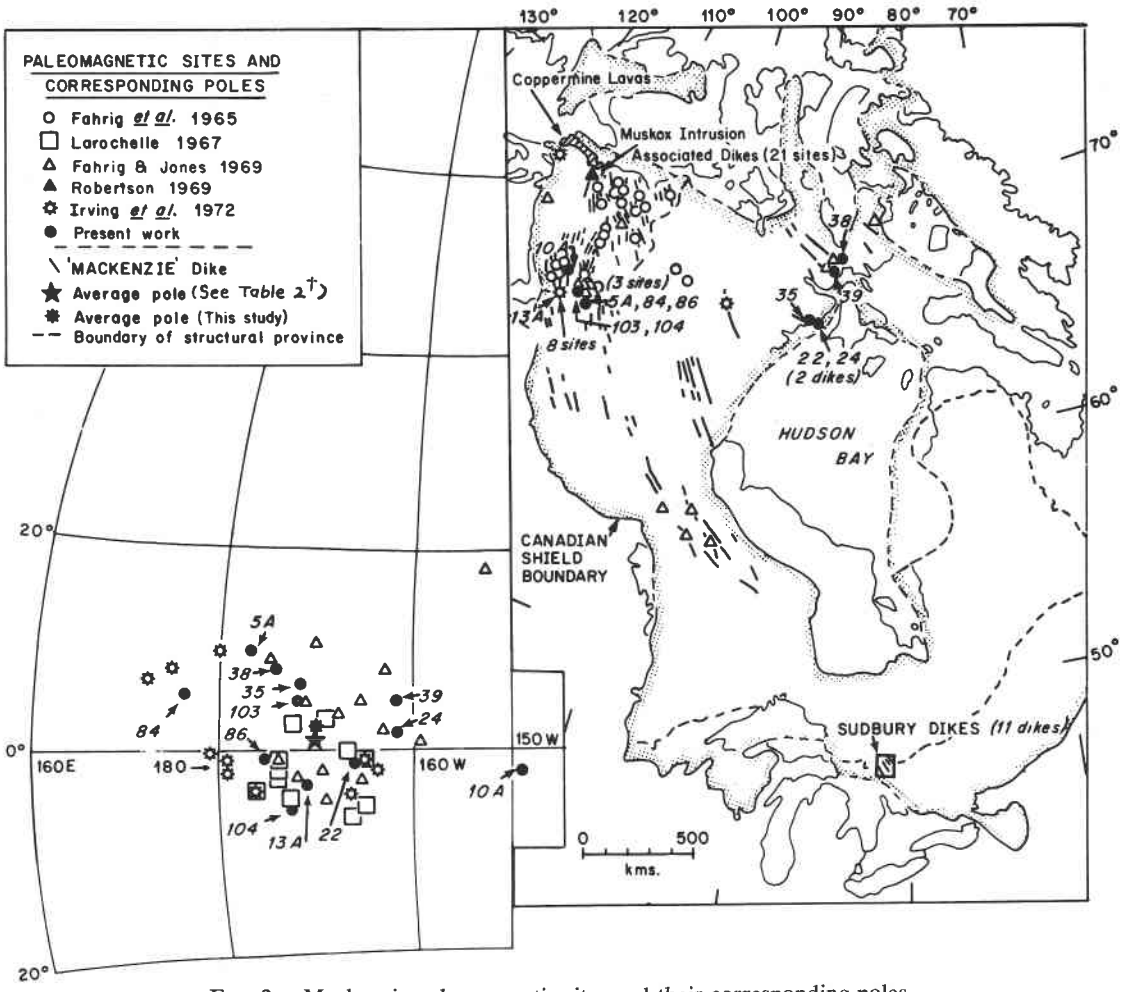



FIG. 2. Mackenzie paleomagnetic sites and their corresponding poles.

TABLE 2. Summary of results from Franklin and Mackenzie diabases

| Description | Pole ($N$, $K$, $A_{95}$) |
|---|---|
| Franklin diabase: previous best estimate* | 05.8°N, 165.8°E (46, 29, 3.8) |
| Franklin diabase: present work | 03.0°N, 161.3°E (3, 181, 9.2) |
| Mackenzie diabase: previous best estimate* | 01.0°N, 171.0°W (35, 78, 3.0) |
| Mackenzie diabase: present work | 02.5°N, 169.8°W (12, 35, 7.4) |

NOTES: $N$ is the number of sites used in the statistical analysis, $K$ is the precision, and $A_{95}$ is the circle of 95% confidence in degrees.
*This pole (Fahrig and Schwarz 1973) includes the 23 Group A sites of Fahrig *et al.* (1971), the 13 sites of Robertson and Baragar (1972), and the 10 sites of Fahrig and Schwarz (1973).
†This pole (Irving *et al.* 1972*b*) includes the sites (11 dikes) of Larochelle (1967), the 14 sites of Fahrig and Jones (1969), and the 10 sites of Irving *et al.* (1972)*b*.

In Table 2, the previous best estimates and the present tentative results of the Franklin and Mackenzie poles are summarized.

## Acknowledgments

I am indebted to D. Halliday of the Gravity Division of our Branch for use of his equipment, facilities, and logistical support, without which this survey would not have been possible; to F. N. Lavery for his valuable field assistance; and to T. Gordon of the Geological Survey of Canada for the use of his helicopter, his camp facilities, and his advice on the Daly Bay geology. I would like to thank E. Irving

for his help and advice throughout this project, and to thank him and J. McGlynn of the Geological Survey of Canada for making available part of their Slave and Bear province collections (1971). One site was also supplied by Irving from his Superior province (1971) collection. The critical reading of an earlier manuscript by J. L. Roy and W. A. Robertson was greatly appreciated.

BRUHNES, B. 1906. Recherches sur le direction d'aimantation des roches volcaniques. J. Phys., **5**, pp. 705–724.

CARD, K. D., DONOVAN, J. F., LOVELL, H. L., LUMBERS, S. B., MEYN, H. D., SAVAGE, W. S., THOMSON, J. E., and THOMPSON, R. 1971. Sudbury–Cobalt Sheet, Ontario. Ontario Dept. Mines and Northern Affairs, Map 2188.

COX, A. 1961. Anomalous remanent magnetization of basalt. U.S. Geol. Surv. Bull., 1083-E, pp. 131–160.

FAHRIG, W. F. and JONES, D. L. 1969. Paleomagnetic evidence for the extent of Mackenzie igneous events. Can. J. Earth Sci., **6**, pp. 679–688.

FAHRIG, W. F. and SCHWARZ, E. J. 1973. Additional paleomagnetic data on the Baffin diabase dikes and a revised Franklin pole. Can. J. Earth Sci., **10**, pp. 576–581.

FAHRIG, W. F., GAUCHIER, E. H., and LAROCHELLE, A. 1965. Paleomagnetism of diabase dykes of the Canadian Shield. Can. J. Earth Sci., **2**, pp. 278–297.

FAHRIG, W. F., IRVING, E., and JACKSON. G. D. 1971. Paleomagnetism of the Franklin diabases. Can. J. Earth Sci., **8**, pp. 455–467.

GRAHAM, K. W. T. 1961. The re-magnetization of a surface outcrop by lightning currents. Geophys. J., **6**, pp. 85–102.

HENDERSON, J. F. 1941*a*. Beaulieu River, Northwest Territories. Geol. Surv. Can., Map 581A.

——— 1941*b*. Gordon Lake South, District of Mackenzie, Northwest Territories. Geol. Surv. Can., Map 645A.

HEYWOOD, W. W. 1967. Geological notes northeastern District of Keewatin and southern Melville Peninsula, District of Franklin, Northwest Territories (parts of 46, 47, 56, 57). Geol. Surv. Can., Pap. 66–40 and Map 14–1966.

IRVING, E., DONALDSON, J. A., and PARK, J. K. 1972*a*. Paleomagnetism of the Western Channel diabase and associated rocks, Northwest Territories. Can. J. Earth Sci., **9**, pp. 960–971.

IRVING, E., PARK, J. K., and MCGLYNN, J. C. 1972*b*. Paleomagnetism of the Et-Then Group and Mackenzie diabase in the Great Slave Lake area. Can. J. Earth Sci., **9**, pp. 744–755.

LAROCHELLE, A. 1967. Paleomagnetism of the Sudbury diabase dyke swarm. Can. J. Earth Sci., **4**, pp. 323–332.

LORD, C. S. 1952. Camsell River, District of Mackenzie, Northwest Territories. Geol. Surv. Can., Map 1014A.

PARK, J. K. 1973. Paleomagnetism of metamorphic rocks of Daly Bay and Melville Peninsula, Churchill Structural Province. Can. J. Earth Sci., **10**, pp. 1079–1088.

ROBERTSON, W. A. 1969. Magnetization directions in the Muskox Intrusion and associated dykes and lavas. Geol. Surv. Can., Bull. 167.

ROBERTSON, W. A. and BARAGAR, W. R. A. 1972. The petrology and paleomagnetism of the Coronation Sills. Can. J. Earth Sci., **9**, pp. 123–140.

ROY, J. L., REYNOLDS, J., and SANDERS, E. 1973. An alternating field demagnetizer for rock magnetism studies. Publ. Earth Phys. Br., **42**, No. 3, pp. 37–45.

ROY, J. L., ROBERTSON, W. A., and KEEPING, C. 1969. Magnetic "field free" spaces for paleomagnetism, rock magnetism and other studies. Can. J. Earth Sci., **6**, pp. 1312–1316.

WANLESS, R. K., STEVENS, R. D., LACHANCE, G. R., and DELABIO, R. N. 1970. Age determinations and geological studies. Geol. Surv. Can., Pap. 69–2A.

WANLESS, R. K., STEVENS, R. D., LACHANCE, G. R., and EDMONDS, C. M. 1967. Age determinations and geological studies. Geol. Surv. Can., Pap. 66–17.

WRIGHT, G. M. 1952. Reliance, Northwest Territories. Geol. Surv. Can., Map 1123A.

——— 1968. Christie Bay, District of Mackenzie, Northwest Territories, Geol. Surv. Can., Map 1122A.

## Appendix

The location, structural, and sampling details of all sites, and the reasons for their assignment to the Franklin (F) and Mackenzie (M) diabases are given. Trends and widths are approximate.

1AD Map 581A (Henderson 1941*b*), southeast corner; width 60 m, tend northeast; sampled at center. Not assigned: may be associated with northwest-trending dikes of Bathurst sheets, but pole is 43° from F pole (Fahrig and Schwarz 1973).

C Baked sediment

5AD Map 1123A (Wright 1952), south shore of Siltaza Lake; width 30 m, trend 165°; sampled at centre. M: part of northwest-trending M dike swarm sampled by Fahrig *et al.* (1965) and Fahrig and Jones (1969); pole within 6° of M pole (Irving *et al.* 1972*b*).

C Granite; sampled from east margin.

10AD Map 645A (Henderson 1941*a*), 62° 52½′N., 113° 17½′W., on west shore of Lee Lake; width 60 m, trend north-northwest; sampled at center. M: belongs to north to north-northwest-

trending M dike swarm sampled by Fahrig *et al.* (1965); pole $<20°$ from M pole, but $\simeq 45°$ from F pole.

13AS Map 1122A (Wright 1968), 62° 13½′N., 11° 28½′W., northwest side of island in Christie Bay of Great Slave Lake. M: belongs to M diabases studied by Fahrig and Jones (1969) and Irving *et al.* (1972*b*).

C Baked syenite; sampled from lower margin out to 0.2 m.

16AD Map 2188 (Card *et al.* 1971), 17.5 km south of Montreal R. bridge in Latchford on hwy. 11; width 40 m, trend northwest; sampled at center. F: parallel to M dikes (Sudbury dikes of Larochelle 1967), but pole within 7° of F pole.

C Pink granite; sampled from margin out to 3.5 m.

22D Map 14–1966 (Heywood 1967) see Fig. 1 of Park (1972); width 15 m, trend 115°; sampled across dike; M: parallel to 38D and 39D; pole within 10° of M pole.

C map unit 5; sampled from northeast margin out to 3.5 m.

24D location—see 22D; width 50 m, trend 120°; sampled across dike. M: see 22D.

31D location—see 22D; width 85 m, trend 130°; sampled from northeast margin to center. F: reversed; pole within 7° of F pole.

C map unit 9b; sampled 0.3 m from northeast margin out to 5 m.

33D location—see 22D, 110 m southwest of 31; width 50 m, trend 130°; sampled from northeast margin to center. F: see 31D.

C map unit 9b; sampled from northeast margin out to 7.5 m.

35AS Map 1014A (Lord 1952), 65° 32′N., 117° 18′W., 10 km north of Lever Lake; sill dipping 30°, trending 45°; from same sill as 37AS. Not assigned: $\simeq 40°$ from F pole.

35D location—see 22D; width 90 m, trend 130°; sampled from northeast margin to center. M: see 22D.

C1 map unit 9a; sampled 1.2 m from northeast margin out to 4 m.

C2 150 m northeast of 35D.

37AS Map 1014A (Lord 1952), 65° 38½′N., 116° 52½′W, $\simeq 6$ km northwest of Wylie Lake; from same sill as 35AS. Not assigned: see 35AS.

38D location—see 22D; width 40 m, trend 100°; sampled from northeast margin to center. M: sampled previously by Fahrig and Jones (1969); K–Ar age of 941 m.y. (Wanless *et al.* 1967).

C map unit 9a; sampled from northeast margin out to 2 m.

39D location—see 22D; width 65 m, trend 130°; sampled from northeast margin to center. M: see 38D; K–Ar ages of 1005 and 1010 m.y. (Wanless *et al.* 1967, 1970).

C map unit 10; sampled 1.5 m from northeast margin out to 5.5 m.

84D Map 1123A (Wright 1952) Sparrow Bay of Nonatcho Lake; width 15 m, trend 165°; sampled at center. M: part of northwest-trending M dike swarm sampled by Fahrig *et al.* (1965) and Fahrig and Jones (1969); pole of adjacent country rocks within 6° of M pole.

C granite gneiss; sampled from margin out to 0.3 m.

86D location—see 84D; trend 160°; sampled at center. M: see 84D.

C granite gneiss; sampled within 0.2 m of margin.

103S Map 1122A (Wright 1968), at shore on southeast corner of Pethei Peninsula across from Fortress Island. M: see 13AS.

104S Map 1122A (Wright 1968), west shore of Kahochella Peninsula on Wildbread Bay, $\simeq 3.75$ km north of Pethei Peninsula; sampled at 3.5 m below upper contact. M: see 13AS.